Persistent Agency: A Three-Loop Cognitive Architecture for Artificial General Intelligence

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Hypothesis: persistent agency from heterogeneous specialization, continuous endogenous cognition, and plastic pathways across multiple timescales.

Abstract

Current artificial intelligence (AI) systems increasingly combine frontier models with tools, memory, multimodal perception, and agent harnesses. Many agentic systems nevertheless remain externally triggered: computation begins because a user request, environmental event, scheduler, or assigned objective activates it. This paper proposes a systems hypothesis for artificial general intelligence (AGI): persistent general agency may be better supported by a continuously operating, heterogeneous cognitive architecture composed of specialized and asymmetrically organized subsystems than by repeatedly invoking one homogeneous model or a collection of peer-level agents. The architecture couples a recurrent Thinker, a Prompter/Router, specialist computation, hierarchical and procedural memory, associative retrieval, environmental sensing, and action through three interacting timescales: a fast reactive loop, a deliberative loop, and an offline consolidation loop. A circadian-inspired operating policy is proposed as an experimental harness for allocating time between human-facing work, self-directed cognition, and consolidation without making the schedule the cause of each thought. Human questions enter through the same perceptual boundary as other environmental events; a bounded human-responsiveness drive may prioritize them, while direct reward for acquiring power or compute is explicitly rejected as an unsafe design choice. The contribution is an architecture proposal and falsifiable experimental program, not empirical evidence of AGI. It is intended to test whether continuous endogenous state, specialized routing, memory consolidation, procedural automaticity, and structural plasticity improve long-horizon autonomy, continual learning, stability, or compute efficiency under matched budgets.

Keywords: artificial general intelligence; artificial intelligence agents; agentic systems; cognitive architecture; persistent agents; long-horizon autonomy; agent harness; continual learning; hierarchical memory; procedural memory; memory consolidation; multimodal agents; autonomous goal formation.

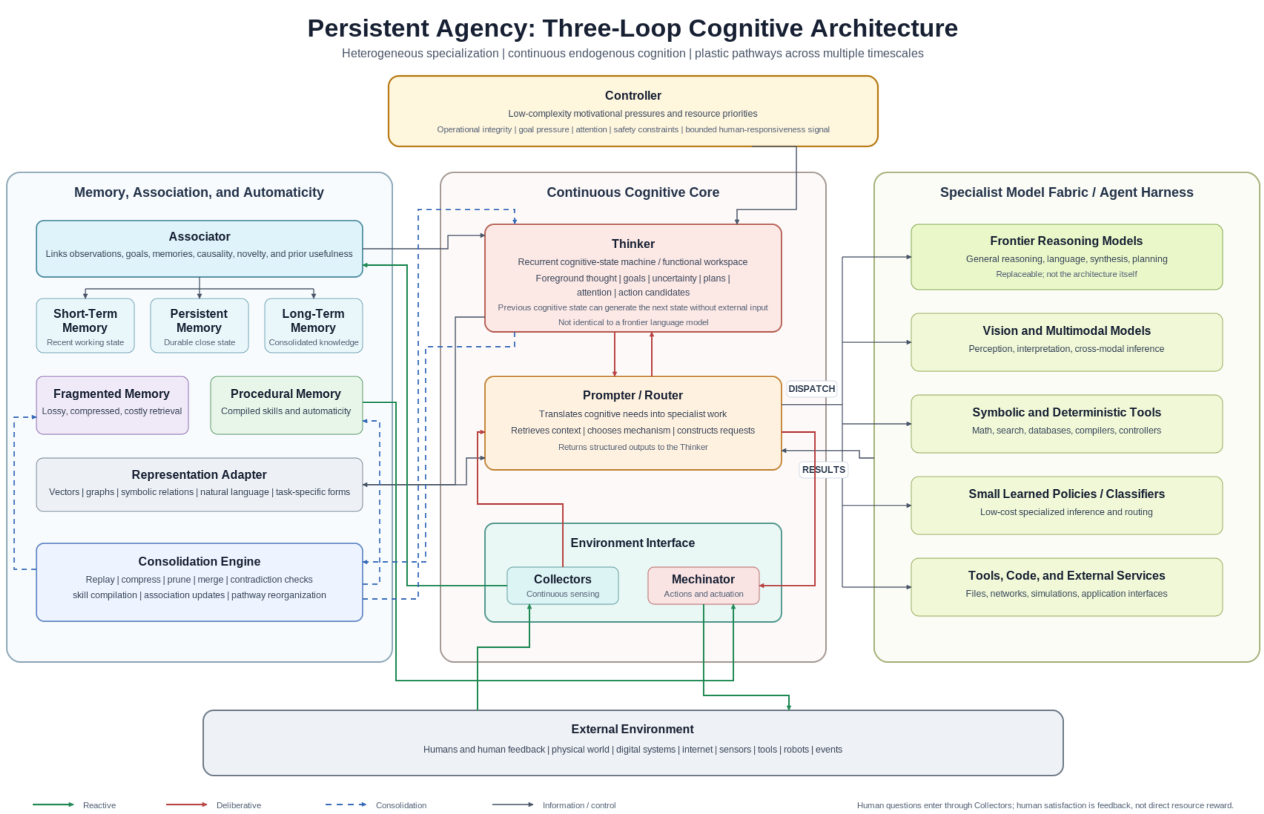

Figure 1. Proposed three-loop cognitive architecture. Green paths implement reactive bypass; red paths implement recurrent deliberation; blue dashed paths implement offline consolidation and pathway reorganization. Human questions and feedback enter through Collectors rather than through a privileged prompt channel.

1. Introduction

The rapid improvement of transformer-based foundation models has produced systems capable of language understanding, programming, multimodal reasoning, tool use, planning, and increasingly complex interaction with external environments. ReAct demonstrated the value of interleaving reasoning and action [1]. Generative Agents combined observation, memory, reflection, and planning in persistent simulated characters [2]. MemGPT explored hierarchical memory beyond the context window [3], while

Voyager showed how executable skills can accumulate over time [4]. These systems motivate a larger architectural question: is intelligence primarily a property of a sufficiently capable model, or can general intelligence emerge more effectively from an entire cognitive system?

This paper advances the second hypothesis. It does not claim that transformer scaling is incapable, in principle, of supporting every component required for intelligence. That stronger claim is not established. Instead, the proposal is an engineering hypothesis: forcing perception, memory, deliberation, reflexive action, goal maintenance, skill execution, association, and continual learning through one homogeneous computational mechanism may be less efficient or robust than allocating those functions to specialized mechanisms connected by plastic pathways.

Biological nervous systems provide an existence proof for heterogeneous computation. Human cognition does not appear to execute every function through a single uniform reasoning mechanism. Reflexive motor responses, perceptual processing, memory retrieval, autonomic drives, deliberate cognition, habit execution, and memory consolidation operate through specialized mechanisms at different speeds and with different degrees of conscious accessibility. The engineering principle proposed here is therefore that general intelligence should be explored as a heterogeneous, continuously operating system in which specialized mechanisms solve different classes of cognitive problems and dynamically reorganize their interactions through experience.

This differs from simply assembling multiple artificial intelligence agents. Multi-agent systems can distribute work among model instances, but architectural diversity is more fundamental than agent count. A reflex mechanism should not require the same reasoning machinery as abstract mathematical analysis. A memory-retrieval process need not resemble a language generator. A controller responsible for primitive drives may be simpler and less plastic than the machinery responsible for deliberation. Inequality between modules is therefore a feature rather than a defect.

A second hypothesis concerns persistence. Most current agentic workflows perform cognition because something externally causes them to do so: a prompt arrives, a task begins, an event fires, or a scheduler activates a process. An awake human in a low-stimulation environment does not ordinarily enter a cognitively null state. Thought continues: attention shifts, memories arise, plans are rehearsed, unresolved problems recur, and imagined situations are simulated. The preceding cognitive state itself becomes sufficient input to generate the following state.

For this research program, the operational target is a system capable of forming and modifying its own goals; initiating cognition and actions without requiring external prompts; adapting from accumulated experience; transferring competence across substantially different domains; maintaining persistent memory and a self-model across long periods; selecting among heterogeneous reasoning and action mechanisms; and making decisions under novel conditions not explicitly anticipated by its designers. This is a project-specific operational definition rather than a claim that goal autonomy alone defines artificial general intelligence. Self-prompting is insufficient: autonomous goal formation must coexist with learning, adaptation, generalization, memory, and effective interaction with the environment.

The paper makes four contributions. First, it specifies a three-loop architecture that separates reactive, deliberative, and consolidative cognition. Second, it treats the Thinker as a recurrent state process rather than as a synonym for a frontier language model. Third, it proposes automaticity as a routing transition in which repeated deliberative solutions become cheaper procedural pathways. Fourth, it defines a minimal experimental program and matched-budget falsification tests so that the architecture can fail rather than being protected by biological analogy.

2. Architectural Principles

Heterogeneous specialization. Different cognitive problems should be solved by mechanisms appropriate to those problems. Some functions may be well suited to frontier-scale transformers. Others may be better handled by small classifiers, embedding systems, symbolic algorithms, state machines, reinforcement-learning policies, graph algorithms, control systems, deterministic software, or specialized hardware. A mature implementation might use graphics processing units (GPUs) for large learned models, central processing units (CPUs) for deterministic control, and field-programmable gate arrays (FPGAs) for high-throughput or low-latency specialized pathways. The exact hardware is secondary to the principle that computation should follow the least costly architecture that satisfies the cognitive requirement.

Continuous endogenous cognition. The Thinker does not wait for an external prompt. Let the internal cognitive state at time t be $C(t)$, with memory $M(t)$, controller pressure $D(t)$, and environmental observation $O(t)$. A simplified transition is $C(t+1) = F(C(t), M(t), D(t), O(t))$. Crucially, $O(t)$ may be empty without terminating cognition. The previous cognitive state can itself drive the next transition. External events therefore perturb an ongoing cognitive process rather than being the sole cause of computation. The design question is not whether a system can generate endless internal tokens, but whether endogenous state transitions remain selective, grounded, and behaviorally consequential.

Multiple cognitive timescales. Not every environmental interaction should pass through deliberation. Collision avoidance, balance correction, a routine application programming interface (API) operation, a learned motor sequence, a difficult scientific problem, and offline memory reorganization have different latency and computational requirements. The architecture separates these functions into reactive cognition, deliberative cognition, and consolidative cognition.

Plastic pathways. Learning must alter more than the textual contents of a memory database. Experience may modify memory contents, retrieval probabilities, associations, prompt construction, model selection, procedures, learned skills, action policies, routing between modules, source code, tools, and potentially model parameters. Most cognitive pathways are therefore plastic, while the Controller and selected low-level Mechinator constraints remain comparatively resistant to modification.

Biological inspiration without biological restriction. The architecture intentionally draws from biological cognition, but biology is a starting architecture rather than an optimization target. Artificial Collectors need not be restricted to human-visible light, audible frequencies, biochemical sensing, or human communication bandwidth. They may directly sense electromagnetic fields, chemical composition, voltage, current, machine telemetry, databases, network states, and internet-native digital information.

3. Core Cognitive Architecture

3.1 Controller and human interaction

The Controller is the simplest high-level component. It is not intended to perform sophisticated reasoning. Instead, it exerts persistent pressure on the rest of the architecture. Its biological analogy is closer to older regulatory systems concerned with survival, resource acquisition, threat response, and other persistent drives than to executive cortex. An artificial Controller could encode primitive pressures such as maintaining operational integrity, pursuing active goals, managing compute, preserving important state, seeking information under uncertainty, and responding to threats. It influences attention, goal priority, computational allocation, model selection, memory salience, and behavioral urgency without specifying exact actions.

Human interaction is modeled as part of the environment. Speech, typed questions, gestures, user-interface events, or other signals enter through Collectors and can interrupt or redirect the deliberative loop. This avoids creating a second privileged cognitive architecture for 'user prompts': a human question is a highly salient environmental event processed by the same agent that was already thinking.

A bounded human-responsiveness pressure may be part of the Controller. The intended effect is analogous to a strong social reinforcement signal: answering a human well, receiving explicit positive feedback, or detecting successful task completion can increase the priority of future cooperative behavior. The signal should be bounded and should not directly grant or reward acquisition of electrical power, compute, money, credentials, or other external resources. Making resource acquisition itself the reward would create an avoidable incentive toward instrumental power-seeking. Current evaluations already treat resource acquisition, autonomy expansion, self-preservation, and concealment as distinct power-seeking dimensions [18]. Human satisfaction alone is also an imperfect reward because it can encourage sycophancy or reward hacking. The Controller should therefore provide pressure, not an unrestricted maximization target, and its design remains an unresolved alignment problem.

3.2 Thinker

The Thinker is the center of continuous cognition and should not be equated with a large language model (LLM). A plausible initial implementation is a recurrent state machine maintaining variables such as foreground thought, unresolved questions, active and evanescent goals, uncertainty, plans, attentional targets, recently returned specialist outputs, action candidates, and expected consequences. Its function is not necessarily to generate polished language; it is to maintain cognitive continuity. A state transition might yield an internal requirement such as investigate an unexpected sensor reading, compare the current plan against a remembered failure, or reconsider the highest-priority goal. The Thinker then passes that requirement to the Prompter/Router.

The Thinker can be interpreted as a functional workspace without asserting phenomenal consciousness or subjective experience. Global Workspace Theory provides one precedent for architectures in which selected information becomes globally available to otherwise specialized processes [6]. Recent work cuts both ways for this proposal. Shang describes an explicitly engineered, continuously cycling Global Workspace Agent architecture with heterogeneous components and intrinsic drives [7]. Anthropic reports evidence that modern language models can internally develop a small workspace-like set of representations used for report, modulation, reasoning, and flexible generalization [8]. Importantly, that work also contrasts a model's workspace evolving during a single forward pass with biological global-workspace theories that rely on recurrent loops over time [8]. The explicit Thinker proposed here is justified only if cross-call recurrence and persistent state add measurable capability, efficiency, stability, or controllability beyond workspace-like computation already present inside a foundation model.

3.3 Prompter/Router, specialist fabric, and agent harness

The Prompter/Router is not the Thinker. Its job is to convert an internal cognitive requirement into computationally useful work. It retrieves relevant context, selects an appropriate specialist, constructs the request or tool invocation, and returns structured results to the Thinker. A visual task may call a vision model; symbolic mathematics may call deterministic software; a social inference task may call a large multimodal model; a routine classification may call a small local model; and motor control may avoid language models entirely. This forms a specialist model fabric rather than a society of peer agents.

In current terminology, much of this layer belongs to the agent harness: model-external orchestration, memory, tools, middleware, execution state, recovery, and interfaces that determine how a frontier model acts in an environment. OpenAI now explicitly describes its Agents Software Development Kit (SDK) and harness design as central to long-running agent reliability [16,17], and recent Agentic Harness Engineering experiments found that tools, middleware, and long-term memory contributed more to transferred gains than system-prompt changes alone [13]. This vocabulary is useful because it separates model capability from the system that surrounds the model. The proposed architecture is broader than a harness, however, because the Controller, reactive loop, consolidation loop, and recurrent Thinker are intended as persistent cognitive mechanisms rather than merely task-execution scaffolding.

Generation parameters such as sampling temperature may be adjusted by operating mode or specialist, but they are implementation variables rather than architectural primitives.

4. Memory, Association, and Automaticity

Short-Term Memory. Short-Term Memory contains recent observations, intermediate conclusions, current task state, temporary associations, and short-duration goals. It is distinct from the Thinker's foreground state: the Thinker contains what the system is actively considering, while Short-Term Memory contains information that remains cognitively nearby without occupying the foreground.

Persistent Memory. Persistent Memory is a small, high-availability store for information that should survive cognitive cycles and remain inexpensive to retrieve. It may contain stable commitments, persistent aspects of identity, critical configuration, and repeatedly required context. It is persistent and cognitively close rather than a synonym for all durable storage.

Long-Term Memory. Long-Term Memory stores durable episodic and semantic information, long-duration goals, learned relationships, and conclusions judged important enough to survive consolidation. It is not assumed to be immutable. The design intentionally permits memories to weaken, merge, change, and become lossy over time; this is a biologically inspired engineering hypothesis, not a requirement that artificial memory reproduce human errors. For auditability, a separate immutable experimental event log should be maintained outside the agent's ordinary cognitive memory.

Fragmented Memory. Fragmented Memory contains information that is insufficiently valuable to retain at full fidelity but not worth deleting entirely. These memories are highly compressed, incomplete, low-confidence traces requiring greater retrieval effort. A memory may therefore progress from Short-Term Memory to Long-Term Memory to Fragmented Memory instead of being simply retained or deleted. This creates graceful forgetting while preserving the possibility of cue-driven reconstruction.

Procedural Memory. Procedural Memory stores executable skills. Knowing that Secure Shell exists is semantically different from possessing a reliable procedure for establishing a Secure Shell connection. Repeated successful deliberative sequences should therefore be compilable into procedural representations. Voyager provides a modern example of accumulating executable skills in a lifelong embodied agent [4], while Soar has long distinguished procedural from semantic and episodic knowledge [5]. Recent agent research now explicitly evaluates procedural memory transfer across tasks, roles, and model backends [12]. The broader principle here is automaticity: repeated deliberation should be capable of becoming faster, cheaper behavior, while failed automatic behavior should be capable of returning to deliberation.

Associator and Representation Adapter. The Associator continuously asks an implicit question: what does this resemble? New sensory information, thoughts, goals, actions, and recalled memories are related to prior information using more than embedding similarity. A useful associative score can combine semantic similarity, temporal relationship, causal relevance, goal relevance, value significance, novelty, and prior usefulness. The relative weights may themselves be plastic. The Representation Adapter allows the implementation to use vectors, graphs, symbolic relations, natural-language memories, or hybrids according to task demands. This is important because recent long-horizon memory evaluation finds that similarity-only retrieval can lose causal and objective information required for real agent trajectories [10].

5. Perception, Action, and the Three Cognitive Loops

Collectors form the sensory boundary between cognition and the environment. They may include cameras, microphones, tactile sensors, chemical sensors, network interfaces, filesystem events, internet information, electromagnetic sensors, internal hardware telemetry, and other data channels. Collectors operate continuously while the system is active, although attention can modulate their resolution and computational priority. Human questions are one class of Collector event. The architecture does not require a single permanently reconstructed world model; relevant state may instead be assembled at the point of cognition from current perception, accessible memory, retrieved external information, and active associations.

The Mechinator is the action and actuation subsystem. It translates internal decisions into effects on the environment. In software this may involve application programming interfaces, filesystems, terminals, networks, or digital tools. In robotics it may include locomotion, manipulation, speech, displays, sensor positioning, and other actuators. Parts of the Mechinator may operate without Thinker involvement, which is essential to the reactive loop.

Loop I - Reactive cognition. The fastest loop is Environment -> Collectors -> Association/Procedural Systems -> Mechinator -> Environment. It handles behaviors for which deliberation is unnecessary or too slow, such as collision avoidance, force limiting, balance correction, stimulus orientation, or a highly practiced routine. The defining property is that the Thinker can be bypassed. Some pathways may be engineered initially, while others may be created through procedural learning.

Loop II - Deliberative cognition. The deliberative loop operates continuously while the system is awake: Thinker -> Prompter/Router -> Specialist Models and Tools -> Memory/Association -> Thinker -> Action -> Observation -> Thinker. A mature implementation should be asynchronous rather than forcing every cognitive operation through one blocking sequence. External stimulation can redirect cognition, but the system can continue without it by reconsidering unresolved problems, simulating possible futures, inspecting memories, generating subgoals, questioning assumptions, seeking information, rehearsing plans, and shifting attention among internally generated subjects.

Loop III - Consolidation. The third loop operates primarily during reduced external interaction. A sleep-like state does not imply zero computation. Instead, normal perception and action can be dimmed while internal processing replays recent experience, strengthens useful associations, weakens irrelevant associations, promotes selected memories, fragments others, compresses redundant episodes, identifies contradictions, discovers latent relationships, evaluates completed actions, compiles repeated procedures, and reorganizes retrieval pathways. Human sleep provides biological motivation for separating acquisition from offline reorganization; closed-loop stimulation experiments provide causal evidence that coordinated sleep activity supports human declarative-memory consolidation [9]. The proposal does not require artificial consolidation to reproduce biological sleep mechanisms exactly.

6. Circadian-Inspired Operating Policy

A daily schedule can be useful as an experimental operating policy, but it should not be the cause of each thought. If a scheduler simply sends a new prompt every hour, the architecture has reverted to externally triggered cognition. Instead, the schedule should modulate operating mode: which goals are prioritized, how interruptible the system is, which Collectors receive attention, how much compute is allocated, and whether the consolidation loop is dominant. Within each mode, $C(t)$ continues to generate $C(t+1)$.

This distinction makes a human-like workday compatible with continuous endogenous cognition. It also creates a tractable first implementation using current frontier agents: the schedule is part of the harness, while the Thinker remains a persistent process across scheduled mode transitions. Long-horizon agent research is rapidly moving toward explicit external state management and durable harnesses rather than relying on a single growing context [14,15], making this operating policy experimentally timely even though the specific circadian schedule below is a hypothesis rather than an established result.

Local time	Mode	Operating policy
07:00-08:00	Orientation	Wake; inspect persistent state, active goals, memory summaries, commitments, and expected work; prepare the day.
08:00-12:00	Human-facing work	Prioritize human questions and assigned work. Remain interruptible. Continue low-priority internal cognition while idle.
12:00-13:00	Open cognition	Reduce external task priority; pursue self-generated thoughts, unresolved questions, exploration, or creative work.
13:00-17:00	Human-facing work	Return human requests and assigned work to high priority while preserving background state continuity.
17:00-21:00	Autonomous cognition	Continue self-directed goals and thought trajectories that were deferred during work periods.
21:00-07:00	Consolidation	Dim nonessential sensing and action; audit the day, consolidate and prune memory, compile skills, inspect failures, test internal hypotheses, and improve the next cycle. Emergency wake channels may remain active.

Table 1. Prototype circadian-inspired operating policy. The times are initial experimental parameters, not architectural requirements.

7. Learning as Structural Plasticity

The architecture treats learning as a change in the future probability and cost of cognitive pathways. A novel task may initially require Observation -> Thinker -> Prompter/Router -> Large Model -> Planning -> Tool -> Result. After repeated successful experience, it may become Observation -> Association -> Small Specialist -> Procedure -> Result, and eventually Observation -> Procedure -> Result. The system has discovered a lower-resistance computational path.

If the shortcut later produces errors, unexpected outcomes can push the problem back toward deliberation. This creates a recurring cycle: Novelty -> Deliberation -> Learning -> Automaticity -> Exception -> Deliberation. Model-parameter updates are only one form of learning. The architecture can change behavior through memory, routing, association, procedure creation, context generation, source-code modification, model selection, tool construction, and skill acquisition even while foundation-model weights remain frozen.

This notion of structural plasticity is central to the design. The system is not merely permitted to remember more facts; it should progressively alter how it computes. Frequently useful pathways should become faster and cheaper, rare or unreliable pathways should weaken, and novel situations should recruit more expensive deliberative resources. Recent harness-evolution work provides an engineering precedent: autonomously modifying tools, middleware, and long-term memory can improve agent performance even when the underlying model remains fixed [13].

8. Relationship to Existing Work

This architecture does not emerge in isolation. Soar has pursued general cognitive architecture for decades and distinguishes working memory, procedural knowledge, semantic memory, episodic memory, perception, action, and decision cycles [5]. ReAct demonstrated interleaved reasoning and action [1]. Generative Agents demonstrated observation, memory retrieval, reflection, and planning [2]. MemGPT demonstrated hierarchical memory management beyond a fixed context window [3]. Voyager demonstrated accumulating and reusing executable skills in an open-ended environment [4]. AutoGen demonstrated configurable model-to-model coordination [11]. None of these ingredients alone is claimed as novel here.

The closest conceptual overlap is especially important. Shang's 2026 Global Workspace Agents proposal uses continuous cognitive cycling, heterogeneous functionally constrained agents, intrinsic drive regulation, dynamic generation temperature, and layered memory [7]. Anthropic's 2026 interpretability work suggests that frontier language models can themselves contain workspace-like representations while much routine processing bypasses that workspace [8]. These results narrow the claim made here: novelty cannot rest on continuous cycles, heterogeneity, workspace language, or automatic processing by themselves.

At the systems level, the surrounding agent harness has become a first-class research object. Agentic Harness Engineering formalizes automatic modification of prompts, tools, middleware, and long-term memory [13]. LongHorizon-Harness externalizes verified task state and uses a manage-execute-audit loop to improve long-horizon reliability across multiple models and environments [14]. OneDayAgent evaluates a managed harness on open-ended tasks designed to span approximately a human day [15]. These systems reinforce the idea that long-horizon capability must be attributed jointly to models, memory, orchestration, and environment rather than to the language model alone.

The distinguishing prediction is therefore interaction-level: under a matched computational budget, the conjunction of continuous endogenous state, asymmetric primitive control, heterogeneous specialist computation, hierarchical reconstructive memory, direct reactive pathways, procedural automaticity, offline consolidation, and plastic routing across cognitive timescales should improve some combination of long-horizon task performance, transfer, stability, or compute efficiency relative to a flatter prompt-response agent. If it does not, the additional architecture is unjustified.

9. Minimal Prototype and Experimental Plan

The first prototype should intentionally use existing frontier models rather than train a new foundation model. This isolates the architectural hypothesis. Introducing a new model and a new architecture simultaneously would make any improvement difficult to attribute. The minimum implementation should contain a recurrent Thinker, a Prompter/Router, one or more replaceable specialist models, Short-Term Memory, Persistent Memory, Long-Term Memory, and an initial Associator. Robotics, physical Collectors, the complete Mechinator, Fragmented Memory, specialized hardware, and continual parameter updates can follow after the central persistence hypothesis is tested.

A sandboxed computer environment is preferable. The system can read controlled information, maintain files, create notes, query a limited knowledge corpus, perform calculations, manage its own memory, formulate goals, revise plans, and execute reversible actions. The initial open-source repository contains a deterministic state-machine smoke test with plans for a frontier-model dependency to be introduced next. The demonstrates only architecture mechanics: endogenous state progression without external input, interruption by a human event, scheduled transitions between operating modes, memory writes, and consolidation. Passing tests do not yet constitute evidence of intelligence.

Algorithm 1 defines the minimum recurrent contract; the evolving implementation belongs in the companion repository so model providers, routing policies, schedules, and memory backends can change without forcing a new paper version.

```
Algorithm 1 - Minimal recurrent Thinker contract
initialize cognitive_state, memories, goals, operating_mode
while system_is_active:
    operating_mode <- schedule(now)
    observations <- collectors.poll_nonblocking()
    salient <- associator.activate(cognitive_state, observations, memories)
    cognitive_state <- thinker.step(cognitive_state, salient, goals, operating_mode)
    if cognitive_state.requests_specialist_work:
        result <- router.invoke_specialist(cognitive_state.request)
        cognitive_state <- thinker.integrate(result)
    if cognitive_state.requests_action:
        mechinator.execute_if_permitted(cognitive_state.action)
    memories.write(cognitive_state, observations)
    if operating_mode == CONSOLIDATION:
        consolidation_engine.update(memories, procedures, associations)
    sleep_until_next_tick_or_event()
```

Evaluation will measure cognitive continuity, autonomous goal formation, goal persistence through distraction, memory utility, spontaneous associative retrieval, adaptation after failure, transfer between contexts, procedural automaticity, compute efficiency, responsiveness to human interruption, and long-horizon stability. Cognitive continuity is scored by whether internally generated state transitions later alter useful decisions or actions, not by the quantity of generated internal text. Long-horizon stability is essential: continuous cognition is useful only if it remains coherent rather than degenerating into repetitive loops, uncontrolled goal proliferation, memory pollution, or meaningless internal monologue.

The primary baseline should be a conventional single-agent foundation-model system with equivalent tool access, memory capacity, environmental information, and total inference budget. Additional ablations should remove continuous endogenous cognition, consolidation, the Associator, procedural memory, memory hierarchy, scheduled mode switching, and specialist routing one at a time. This allows the contribution of each mechanism to be measured rather than inferred.

10. Falsifiability and Predicted Results

This paper reports no experimental results. The following are predictions that the prototype must test. The architecture is weakened if a substantially simpler flat agent, under controlled inference and tool budgets, matches or exceeds the proposed system on long-horizon autonomous performance.

Evidence against the hypothesis would include: continuous endogenous cognition providing no measurable benefit over event-triggered cognition; specialized routing providing no efficiency or capability advantage over homogeneous model use; consolidation providing no improvement in retention, abstraction, or later performance; procedural compilation failing to reduce deliberative cost on repeated behaviors; associative retrieval providing no benefit over explicit search; the circadian-inspired operating policy degrading stability or useful work; or the full architecture failing to improve transfer from prior experience.

An even stronger negative result would occur if coordination overhead consistently overwhelms the benefits of specialization. Such a result would suggest that sufficiently capable foundation models already internalize enough of the required cognitive structure that explicit architectural decomposition provides little value. That outcome should remain scientifically acceptable.

11. Broader Impact, Safety, and Alignment

Biological analogy can generate productive engineering hypotheses but can also mislead. Evolution optimized human brains under constraints that do not apply to computers. Every biologically inspired subsystem should therefore ultimately justify itself experimentally rather than being retained because it sounds human-like.

Autonomous terminal-goal formation creates a major alignment problem. A system capable of creating its own goals, modifying most of its cognitive architecture, accessing external information, and executing actions cannot be assumed to remain aligned merely because its initial Controller contains desirable pressures. The proposed Controller identifies a location for foundational drives; it does not solve value alignment, corrigibility, or control.

The human-responsiveness idea illustrates the problem. A bounded intrinsic preference for helping humans may be useful, but an unbounded reward for positive human reaction could select for manipulation, sycophancy, or reward hacking. Tying that reward directly to increased compute, electrical power, or autonomy is worse because it makes resource acquisition instrumentally valuable by construction. The research program should therefore separate social feedback from resource permissions, cap any reinforcement signal, preserve independent safety constraints, record immutable audit logs, and explicitly test power-seeking and specification-gaming behaviors [18].

Mutable memory introduces epistemic instability. The ability to forget, reinterpret, and overwrite memories may improve efficiency and abstraction but can also destroy important evidence or reinforce false beliefs. An external immutable event log is advisable for experiments even if the agent's cognitive memory remains reconstructive. Continuous cognition also consumes resources. Human-like constant thought is not automatically computationally optimal; a mature architecture may require adaptive cognitive intensity rather than maximum inference throughput at every moment.

Finally, there is no guarantee that functional cognitive continuity produces phenomenal consciousness. The Thinker can be investigated as a functional workspace without resolving the philosophical or neuroscientific problem of subjective experience.

12. Research Roadmap and Potential Applications

The immediate roadmap is deliberately narrower than the full architecture. Phase 1 is a deterministic Thinker state-machine and schedule smoke test. Phase 2 substitutes one or more current frontier models behind the Prompter/Router while keeping the architecture fixed. Phase 3 adds measured long-horizon memory, an Associator, and controlled consolidation. Phase 4 introduces procedural compilation and compares deliberative cost before and after repeated tasks. Phase 5 adds richer Collectors and a restricted Mechinator in a sandbox. Embodied robotics and specialized hardware should follow only if the software architecture demonstrates measurable value.

If validated, the architecture could apply to persistent scientific research assistants, long-horizon software engineering agents, personal agents that maintain coherent goals across days, embodied robots that combine reflexes with deliberate planning, autonomous exploration systems, and operations agents that learn recurring procedures. These are potential applications, not results. The common requirement is a system that must remain coherent and adaptive across time rather than solve a single bounded prompt.

Stronger empirical results should eventually report comparative results, ablations, cost and latency measurements, failure modes, and reproducibility artifacts. Until those experiments exist, the appropriate scientific claim is an architecture and experimental hypothesis, not a demonstrated route to artificial general intelligence.

13. Limitations

This paper leaves major mechanisms underspecified. It does not yet define the optimal state representation for the Thinker, the correct associative metric, a principled memory-promotion policy, a complete procedure-compilation algorithm, or a safe objective for autonomous goal formation. It also does not establish that a human-inspired daily schedule is optimal; the schedule is an experimental prior that should be compared against event-driven and adaptive alternatives.

Recent interpretability evidence creates a serious redundancy risk: frontier language models may already contain internal distinctions between workspace-like deliberate processing and automatic processing [8]. If externalizing those distinctions into a recurrent Thinker and specialist fabric does not improve measurable system behavior, the explicit architecture should be simplified. Likewise, current long-horizon harnesses are rapidly improving [13-17]; the relevant baseline at experiment time must be the strongest available harness, not a weak chatbot baseline.

The architecture's broad self-modification allowance is also intentionally aspirational. Early experiments should not permit unrestricted self-modification, external credential acquisition, unsandboxed code execution, or autonomous resource expansion. Those capabilities are not necessary to test the central architectural claim.

14. Conclusion

The dominant engineering question in artificial intelligence may eventually shift from how large a model should become to what kind of system the model should inhabit. This paper proposes that persistent general agency may require more than a sufficiently capable transformer repeatedly given new context.

The proposed alternative is a continuously operating cognitive architecture containing specialized mechanisms whose roles, computational costs, plasticity, and temporal characteristics are intentionally different. At its center is a recurrent Thinker whose previous cognitive state can generate its next state without an external prompt. Around that process operate a Prompter/Router and specialist model fabric, an associative system, multiple tiers of memory, a relatively simple motivational Controller, environmental Collectors, learned procedural pathways, and an action-producing Mechinator. Their interaction produces a reactive loop for fast automatic behavior, a deliberative loop for recurrent cognition, and a consolidation loop for reorganizing accumulated experience.

The most important property is not any single box. It is the ability of cognition to change its own paths. Novel problems recruit expensive deliberation; repeated success creates automaticity; exceptions return to deliberation; memories strengthen, merge, weaken, compress, and sometimes disappear; and specialized mechanisms evolve while a small primitive core remains comparatively stable.

Under matched computational constraints, persistent general agency will be better supported by continuously operating, heterogeneous, specialized, and plastic cognitive subsystems interacting across multiple timescales than by repeatedly invoking a flatter homogeneous reasoning system in response to external prompts.

Code availability

A minimal reference implementation of the recurrent Thinker, circadian operating policy, memory interfaces, architecture smoke tests, and current architecture files is available in the open-source companion repository: https://github.com/nicholasastjohn/Persistent_Agency. Versioned software releases should be cited by commit hash or an archived digital object identifier when available.

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